

Instrumentation Troubleshooting & Loop Diagnostic Reference Guide

Field Manual for Loops, Transmitters & Control Valves

EXECUTIVE OVERVIEW & CORE DIAGNOSTIC PROMISE

"Most instrumentation courses teach you what a transmitter is. This guide and course teach you what to do when it lies to you."

Why This Matters: This reference guide enables field technicians to evaluate abnormal 4–20 mA signals, choose the appropriate measurement sequence, isolate root causes without replacing operational equipment, and document defensible diagnoses before standing in front of a live, operating loop.

1. Executive Summary & System Overview

- **Course Title:** Instrumentation Troubleshooting: Loops, Transmitters & Valves
- **Companion Sim:** Loop Troubleshooting Sim (re-themed from Site Safety Sim architecture)
- **Target Platforms:** Udemy (launch platform), Teachable (scaling platform)

Course Structure: \$74.99 list price · Intermediate Level · 7 sections · 68 lectures · ~9h 20m total run time

Strategic Shift & Market Alignment

- **Demand Proven:** 51 courses in Udemy control engineering; bestsellers pricing at \$79.99 (16.5h) and \$84.99 (12.5h).
- **Troubleshooting Slot Gap:** Closest competitor has 6,054 students, 607 reviews, and a 4.3 rating, but its syllabus is a list of device types rather than a structured diagnostic method.
- **Fundamentals Slot:** Beginner 4–20 mA courses sell at \$12.99 with 2,351 students over 3.5h.
- **Industry Pain Point:** Technical training is criticized as "information delivery, not skill building" — hands-on practical troubleshooting matters more than brand power.
- **The Product Wedge:** A fault-first course where every module opens with a symptom, walks the measurement sequence, and presents the physical evidence that closes the call. The Sim provides the interactive practice tool competitors lack.

2. Six-Step Diagnostic Sequence & Live-Zero Signal Reference

Procedural Diagnostic Sequence

Where sequence affects safe and correct work, observe the exact six-step isolation process:

1. **Define symptom** — Identify the specific reading anomaly or process discrepancy.
2. **Verify safety** — Confirm safety boundaries, permits, and LOTO requirements before taking measurements.
3. **Check signal path** — Trace power, wiring, terminal blocks, and ground path connections.
4. **Split the loop** — Perform split-half isolation to determine whether the fault is in the field device, field wiring, power supply, or controller/card.
5. **Prove root cause** — Apply physical tests (meter measurements, calibration checks, stroke tests) to confirm the specific failure mode.
6. **Document** — Record as-found/as-left values, complete the case log, and verify with a final loop check.

Live-Zero & NAMUR Signal Levels Reference

Signal Level	Status / Condition	Diagnostic Interpretation
0.0 mA	Open Circuit	Broken wire, disconnected terminal, or dead power supply. Does not indicate a dead transmitter alone.
3.6 mA	Low Failure (NAMUR NE 43)	Transmitter sensor fault / internal alarm driven low.
3.8 mA	Under-Range	Process signal below lower range value (LRV).
4.0 mA – 20.0 mA	Valid Process Signal	Normal operating range (4.0 mA = 0%, 20.0 mA = 100%).
20.5 mA	Over-Range	Process signal above upper range value (URV).
21.0 mA	High Failure (NAMUR NE 43)	Transmitter sensor fault / internal alarm driven high.

3. Tank Level Condition & Diagnostic Comparison Table

Scenario / Condition	Physical Tank Condition / Level	Transmitter Reading	Diagnosis / Root Cause
Normal Empty Tank	0% (Empty)	4.0 mA (0%)	Normal calibrated dry/wet leg balance.
False High Level (Lost Wet Leg)	0% (Empty)	47%	Lost wet leg fill fluid; hydrostatic head reference lost, causing transmitter to read false high level.

4. The Seven Filmed Field Faults & Capstone

#	Fault Name	Section	Sim Scenario	Key Diagnostic Lesson
1	Open loop vs dead transmitter	2	"The Dead Signal"	0 mA indicates an open circuit, not a dead transmitter.
2	Ground loop and induced noise	2	"The Noisy Loop"	Shield must be grounded at one end only.
3	False high level from lost wet leg	3	"The False High Level"	Loss of wet leg fluid causes transmitter to read high, not low.
4	Temperature reads ambient	3	"The Ambient Thermocouple"	Short near junction box creates a new measuring junction.
5	Intermittent trip that isn't the switch	4	"The Phantom Trip"	Verify first-out event and high-resolution trending.
6	Valve at 50% signal, 0% travel	5	"The Stuck Valve"	Verify signal at positioner before pulling valve trim.
7	Reading jumps to full scale intermittently	6	"The Upscale Jump"	Intermittent open circuit in sensor circuit.

Capstone Scenario: PIC-510 pressure loop "hunting" — cause is an undersized or plugged air-supply regulator, not controller tuning parameters.

5. Platform Economics & Business Model Matrix

Platform / Channel	Revenue Keep	Operational Notes & Strategy
Udemy (Instructor Coupon/Referral)	97% of net	Drive ALL owned traffic through custom referral links. Represents a 2.6x revenue difference compared to organic sales.
Udemy (Organic Marketplace / Ads)	37% of net	Default acquisition path for organic marketplace browse and ad campaigns.
Udemy Business Subscription Pool	~15%	Share based on consumption time. Adjusted downward from prior earnings levels in Jan 2026.
Teachable Starter	\$39/mo (\$29/mo ann.)	7.5% transaction fee applied to sales.
Teachable Builder	\$89/mo (\$69/mo ann.)	0% course transaction fee (standard payment processing fees apply via teachable:pay).
Teachable Growth	\$189/mo (\$139/mo ann.)	0% course transaction fee (standard processing fees apply).

6. Two-Track Architecture & Curriculum Design

Track A: Premium Video Course Structure

- **Terminal Outcome:** Isolate any unfamiliar instrument fault to field device, wiring, power, or controller using a documented measurement sequence, and prove the fix with a loop check.
- **Curriculum Metrics:** 7 sections · 68 lectures · ~9h 20m total runtime · 4 quizzes · 3 decision drills · 1 capstone.
- **Hard Rule:** No lecture over 9 minutes duration.

Section Title	Lecture s	Runtim e	Core Focus & Deliverables
1 — How to Diagnose Anything	7	42m	5-question triage, split-half isolation, safety boundaries, evidence trail. Quiz 1 (8 questions).
2 — The 4–20 mA Loop	11	78m	Loop power, 2/3/4-wire, 4 measurement methods, live-zero diagnostics, 2 filmed faults. Decision Drill 1 (5 questions).

Section Title	Lectures	Runtime	Core Focus & Deliverables
3 — Transmitters	14	2h 05m	Calibration vs ranging, 5-point as-found, DP level, radar, DP flow, RTD, thermocouple, 2 filmed faults. Quiz 2 (12 questions).
4 — Switches & Discrete I/O	9	62m	NO/NC, fail-safe, setpoint/deadband, interlock proving, 1 filmed fault. Quiz 3 (10 questions).
5 — Control Valves	13	1h 48m	Air supply, fail direction, stroke test, positioner calibration, hunting diagnosis, 1 filmed fault. Decision Drill 2 (6 questions).
6 — Controller & Loop Faults	9	64m	P&ID reading, loop diagrams, scaling, filtering, alarm first-out, 1 filmed fault. Quiz 4 (10 questions).
7 — Case Studies & Capstone	7	61m	3 full cases, documentation, personal playbook, filmed capstone. Decision Drill 3 (5 questions) + Capstone.

Track B: Loop Troubleshooting Sim Mapping

Site Safety Sim Module	Loop Troubleshooting Sim Module	Content Source
Plant Map (zones)	Loop Map — trace signal chain: Process → Sensor → Transmitter → Wiring → Card → Controller → Display.	Five-Question Triage Card + Split-Half Worksheet
Permit Panic (sort cards)	Band Sort — sort falling readings into: Open Circuit, Low Failure, Under-Range, Valid, Over-Range, High Failure, Short/Foreign Source	Form 3 (4–20 mA Reference Sheet)
Scenario Simulator	Fault Simulator — 7 filmed faults as interactive scenarios with fail states	Section 7 cases + Decision Drills 1–3
Safety Vault (cloze cards)	Formula Vault — fill-in-the-blank for mA↔%↔EU conversion, alarm bands, tolerance math, hysteresis	Form 3 + Calculators.xlsx

Site Safety Sim Module	Loop Troubleshooting Sim Module	Content Source
Boss Fight (rapid fire)	Quick Draw — 60-second rapid-fire diagnostic calls	Quiz bank + Master Checklists

7. Shared Asset Pack & Toolkit

The Template Pack (Forms 1–7)

Form	Purpose	Used in Sim as
1. Five-Question Triage Card	Pre-work triage	Loop Map intro + quiz logic
2. Split-Half Isolation Worksheet	Isolation log	Loop Map progression tracker
3. 4–20 mA Reference & Conversion Sheet	Signal interpretation	Band Sort reference + Formula Vault
4. Loop Check Form (Cold and Hot)	Loop verification	Loop Map completion checklist
5. As-Found / As-Left Calibration Record	Calibration documentation	Simulator debrief output
6. Case Study Worksheet	Fault documentation	Fault Simulator scenario template
7. Master Diagnostic Checklists by Symptom	Symptom-based troubleshooting	Quick Draw question source

The Calculator Workbook (5 Tabs)

Tab	Purpose	Used in Sim as
Start Here	Instructions + formulas	Onboarding screen
Cal Record	As-found/as-left with PASS/FAIL	Simulator debrief auto-population
Loop Check	Five-point hot loop check	Loop Map verification
mA Converter	mA↔%↔EU with NAMUR bands	Band Sort answer key

Tab	Purpose	Used in Sim as
Split-Half Log	Isolation step tracking	Loop Map progression + flags

8. Course Landing Page Specifications

Title & Subtitle

- **Title:** Instrumentation Troubleshooting: Loops, Transmitters & Valves
- **Subtitle:** Diagnose 4–20 mA loops, transmitters, switches, and control valves with a repeatable measurement sequence — filmed on real equipment.

"What You'll Learn" Objectives

1. Isolate any instrument fault to field device, wiring, power supply, or controller using a documented split-half sequence.
2. Measure and interpret a live 4–20 mA loop four different ways without breaking the process.
3. Perform and document a five-point as-found / as-left calibration on pressure, level, flow, and temperature transmitters.
4. Diagnose control valve, positioner, solenoid, and actuator faults — and tell mechanical problems apart from tuning problems.

Requirements & Audience

- **Requirements:** Basic DC electrical knowledge (voltage, current, series circuits); multimeter access helpful but not required; no PLC programming experience needed.
- **Target Audience:** New E&I / I&C technicians; electricians and maintenance techs moving into instrumentation; operators and engineers; apprentices preparing for plant troubleshooting evaluations.

9. Safety Boundaries & Complaint-Driven Quality Rules

SAFETY MANDATES & BOUNDARIES

- **Bench-Footage Safety Pass:** Every filmed fault must show permit/LOTO/procedure steps, or explicitly state it is a bench setup.
- **Privacy & Signage:** Blur tags, badges, and site signage on all bench and field footage.

- **Site Authorization:** Obtain explicit site authorization before bringing cameras near live plant equipment.
- **Zero Bypass Policy:** Never bypass LOTO, work permits, or alarm management procedures.

Complaint-Driven Quality Rules

Learner Complaint	Mandatory Quality Rule / Countermeasure
Muffled / Echoey Audio	Use dynamic mic close-in, treat room acoustics, and loudness-normalize every lecture to the same target level.
Hard to Understand	Upload fully corrected captions manually.
Missing Hands-on Labs	End every section with an active task or fault call the student performs, not just a video watched.
Outdated Content	Date-stamp course, post visible annual update notes, and avoid software screens locked to specific minor versions.
Course Padding	Maintain a strict 9-minute lecture cap; eliminate recap-of-the-recap lectures and setup fumbling.
Dry / Abstract Material	Open every module with an observable physical symptom before presenting theoretical principles.
Vendor Lock-in	Teach diagnostic methods vendor-neutrally; label proprietary screens as examples only.
Low Completion Rates	Keep lectures short, provide visible progress checkpoints, and offer downloadable artifacts at each section end.
Information Delivery vs Skill Building	Treat filmed faults as the non-negotiable core product; never cut faults to add theory lectures.

10. Technical Tool Stack & Hardware Minimums

- **Software Tools:** Cursor, OBS Studio, Audacity, DaVinci Resolve, Canva, HandBrake, Netlify Drop, Obsidian, Marker, Playwright.
- **Hardware Minimums (\$145–\$430 Total Budget):**
 - USB Dynamic Microphone (Shure MV7 or FIFINE AmpliGame): \$80–\$250
 - Boom Arm & Pop Filter: \$20–\$40
 - Soft LED Panel: \$30–\$100
 - Bench Camera Mount / Overhead Arm: \$15–\$40
 - Webcam / Phone for presenter shots: \$0

11. Confirmation Requirements

Needs confirmation: The specific calendar upload and launch dates for post-launch Sections 4–7 on the dated release schedule require confirmation prior to publishing the course landing page.

Market Validation Status: Market-validated. Proceed to build.